TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Study of Fabric Defects

Reading cloth backwards: the defect atlas of the weaving shed, each fault's appearance, its mechanism cause, and its cure — closing with the four-point grading system that converts defects into a commercial verdict.

FM-4102 · ADVANCED WEAVING TECHNOLOGY-II SESSIONAL · EXPERIMENT 3 · FABRIC MANUFACTURING SPECIALIZATION

1. Aim and theory

Aim: to identify the principal woven-fabric defects on grey cloth samples, assign each to its cause in preparation, loom mechanism or material, state the remedy, and grade a roll by the four-point system.

A defect is a mechanism's signature written in cloth. The classification used in the shed — and in this experiment — is by direction and origin:

- **Warp-way defects** run along the cloth: their cause travels with a warp end or a warp-direction machine element (reed, heald, beam).
- **Weft-way defects** run across the cloth: their cause is an event in time — one pick or one loom moment, repeated or single.
- **General/area defects** cover patches or the whole piece: preparation, dyeing-adjacent or handling causes.

The diagnostic rule follows at once: *a defect that repeats at a fixed interval repeats with the period of some rotating part or some repeating event* — measure the interval and you have halved the search (a repeat every cloth-roll circumference points at the take-up roller; every gear circumference at that gear).

2. Apparatus and samples

- Grey fabric samples (loom-state and inspected rolls) showing the atlas defects; inspection frame/table with transmitted and reflected light; measuring tape and ruler; pick glass (linen tester); scissors and marking chalk; defect atlas/reference photographs; four-point grading pad.

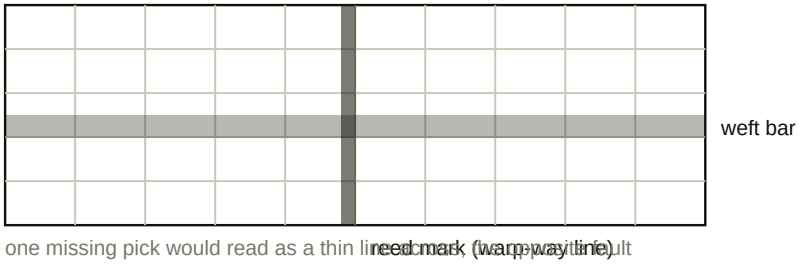


Figure 1 — Two signatures in one sketch: a **weft bar** (a band of different pick density/appearance running across — cause is an event in time: let-off or take-up disturbance) and a **reed mark** (a warp-way streak — cause travels with the reed/warp path). Direction alone assigns each to its family.

3. The defect atlas (identification standard for this experiment)

DEFECT	APPEARANCE	PRINCIPAL CAUSES	REMEDY / PREVENTION
Missing pick	Thin line across the cloth; gap in weft cover	Weft break not detected; insertion fault; empty feeder/package	Maintain weft stop motion & feeder; loom patrol
Double pick	Thick line across; two picks in one shed	Broken pick not removed before restart; insertion overlap	Pick-finding discipline at restarts; auto pick repair where fitted
Weft bar	Band across cloth of visibly different density/shade	Let-off tension change, take-up slip/backlash (Expt 2), loom stoppage mark	Service let-off/take-up; stop-mark procedure (fell re-positioning)
Thin / thick place (weft)	Short across-band of lower/higher cover	Weft count variation, feeder tension variation	Weft lot control; feeder maintenance
Broken end / missing end	Warp-way line of missing cover, may run metres	Warp break with failed stop motion; end out from warping	Warp stop motion upkeep; beam quality; gaiting checks
Reed mark	Fine warp-way streaks, often periodic across width	Damaged/misaligned reed dents; wrong denting	Reed inspection and replacement; correct denting plan
Slack / tight end	One end weaving wavy (slack) or shining/taut (tight), warp-way	Beam tension variation, sticky size, end tangled at lease	Warping/sizing quality; beam storage; lease care

Temple mark	Distorted/damaged band near selvedges	Temple set wrong, rings worn, width mismatch	Temple setting to cloth width; ring renewal
Selvedge defects	Wavy, torn, curled or thick edges	Leno/tuck-in device faults (Expt 1), weft tail length, edge tension	Service selvedge devices; cutter settings
Oil / stain	Localised marks, often with machine geography	Lubricant fling, handling, dirty shuttles/rapiers	Controlled lubrication; clean handling; guards
Snarl / kinky weft	Small loops of weft doubled into the cloth	Over-twisted lively weft, insertion tension spikes	Weft conditioning (steaming), feeder tension control
Hole / tear	Local rupture	Foreign matter through the reed, mechanical damage, harsh beat-up on a fault	Shed housekeeping; gentle restart after faults

4. Procedure

1. Examine each sample on the inspection frame in reflected, then transmitted, light. Record: defect name, direction, dimensions, position, and whether it repeats — with the repeat interval measured.
2. With the pick glass, verify the structural story (picks present/absent, ends missing, dent spacing) before naming the cause — appearance suggests, structure proves.
3. For each defect, write the cause chain to a mechanism or preparation stage studied in Experiments 1–2, and the remedy at that stage.
4. Grade the provided roll by the four-point system (Section 5) and give the commercial verdict.

5. Worked calculation — four-point grading

GRADING A ROLL (MODEL DATA)

Rule: defects are penalised by length — up to 3″: 1 point; 3–6″: 2; 6–9″: 3; over 9″ or any hole: 4 — with a maximum of 4 points per linear yard, and the result normalised:

Points per 100 square yards = (total penalty points × 3600) ÷ (roll length in yards × width in inches).

Roll: 120 yards long, 60 inches wide; inspection total = 90 penalty points.

Points/100 yd² = 90 × 3600 ÷ (120 × 60) = 324,000 ÷ 7,200 = **45 points**.

Against the common acceptance limit of 40 points/100 yd² for first quality, this roll is graded

> **second quality** — the calculation, not the inspector's mood, delivers the verdict; the defect-by-defect record then tells the mill *why*.

6. Precautions

- Inspect in consistent lighting; a defect graded under one light and disputed under another wastes everyone's time — the frame's conditions are part of the method.
- Measure repeat intervals before cutting samples; a cut destroys the periodicity evidence.
- Do not name a cause from appearance alone where structure can be checked (pick glass first).
- Grade the whole roll length provided — sampling the good middle is the oldest fraud in inspection.

7. Practice (viva preparation)

1. A defect repeats every 1.26 m of cloth. The take-up (cloth) roller is 40 cm in diameter. Make the diagnosis argument.
2. Distinguish weft bar from thin place, and missing pick from broken pick, in appearance and cause.
3. A roll 100 yards × 45 inches collects 60 penalty points. Grade it against the 40-point limit.
4. Why is a reed mark warp-way while a stoppage mark is weft-way?
5. An oily stain appears at random positions but always near one selvedge. Where do you look first?

Answer key

1. Roller circumference = π × 40 = 125.7 cm ≈ 1.26 m — the defect's period equals the cloth roller's circumference, so something on that roller (a damaged patch, a hard lump, eccentricity) marks the cloth once per revolution. Period matching converts a mystery into an address.
2. **Weft bar:** a band of altered pick density/appearance from a tension/take-up event; picks are present but wrongly spaced. **Thin place:** a short band of reduced cover from the weft itself (count/tension variation). **Missing pick:** a pick absent (insertion failure or undetected break). **Broken pick:** a pick inserted partly and broken — visible as a partial line ending mid-cloth, usually at the receiving side.
3. Points/100 yd^2 = 60 × 3600 ÷ (100 × 45) = 216,000 ÷ 4,500 = **48 points** → above 40, second quality.
4. A reed fault is attached to fixed positions in the warp path, so it prints continuously along the warp direction. A stoppage is an event at one moment of production: the fell relaxes or the restart misplaces picks across the whole width at that instant — a line across the cloth, i.e. weft-way.
5. At that selvedge's machinery: the leno/selvedge device, its lubrication point, the temple on that side, and the weft cutter — random in time but fixed in geography means a local intermittent source, most often lubricant fling from the nearest moving part.